Q1.

a) $A = \{ x \in N : 2x^2 + 3x - 2 = 0 \}$

Sol: $2x^2 + 3x - 2 = (2x-1)(x+2) = 0$

$x = \frac{1}{2}$ or $x = -2$

Since $x \notin N$, A is an empty/null set, denoted by $A = \{\}$ or $A = \emptyset$
Cardinality of A is zero or $|A| = 0$

b) $B = \{ x \in Z : x^2 < 10 \}$; $x^2 < 10 \Rightarrow -\sqrt{10} < x < \sqrt{10}$

$B = \{0, 1, -1, 2, -2, 3, -3\}$
$|B| = 7$

c) $C = \{ x \in R : x^2 + 2x + 3 = 0 \}$

Sol: $x^2 + 2x + 3 = 0$ does not have any real solutions.
$\therefore$ $x \notin R$ and C is an empty set, i.e $C = \{\}$ and $|C| = 0$

d) $D = \{-1, \{0,1\}, 2, 3, 4, \{5\}, 6, \{7,8\}\} \cap \{x \in Z : |x-3| \geq 2\}$

$\{x \in Z : |x-3| \geq 2\} = (-\infty, 1] \cup [5, \infty]$

$\therefore D = \{-1, 6\}$ and $|D| = 2$

Q5. $A = \{1, 2, 3, 4, 5\}$ and $B = \{1, 3, 5, 7, 9\}$

a) $\{ b \in B : 2b \in A \} = \{1\}$

b) $\{ a \in A : a^2 \in B \} = \{1, 3\}$

c) $\{(a,b) \in A \times B : a+b \text{ is odd}\} = \{(2,1), (2,3), (2,5), (2,7), (2,9),$
$(4,1), (4,3), (4,5), (4,7),$
$(4,9)\}$

$(a+b)$ is odd only when one term is even and other is odd.
Since, B only contains odd numbers, we take evens from A $(2, 4)$ and make pairs such that $a \in \{2,4\}$ and $b \in B$

Q6.

a) $\emptyset \subseteq \{R\}$

→ True, because null set is a subset of every set.

b) $R \subseteq \{R\}$

→ False, because elements of R are numbers whereas element of $\{R\}$ is a single set. For example, $\{1\} \nsubseteq \{\{1\}\}$, rather $\{1\} \in \{\{1\}\}$

c) $\{R\} \subseteq R$

→ False. For example, $\{\{1\}\} \nsubseteq \{1\}$. Same reason as above. $\{R\}$ has one element which is a set of all real numbers whereas R is a infinite set of real numbers.

d) $\{Z\} \subseteq \{R\}$

→ False, because both are singletons with differents elements. The correct statement would be $Z \subseteq R$

Q8. $A \subseteq P(A)$

Sol:

$$\boxed{\begin{array}{l} A = \{\phi\}, \quad P(A) = \{\phi, \{\phi\}\} \\ \phi \in A \text{ and } \phi \in P(A) \\ \therefore A \subseteq P(A) \end{array}}$$

Q9.

a) $A = \{S \in P(N) : S \text{ is an infinite set}\}$

$$\boxed{N, \{x \in N : x \text{ is odd}\} \in A}$$

b) $B = \{S \in P(N) : S \text{ and } N \setminus S \text{ are infinite sets}\}$

$$\boxed{\{x \in N : x \text{ is odd}\}, \{x \in N : x \text{ is even}\} \in B}$$

Q10) $A = \{1, 2, 3, 4\}$

$P(A) = \{\phi, \{1\}, \{2\}, \{3\}, \{4\}, \{1,2\}, \{1,3\}, \{1,4\}, \{2,3\}, \{2,4\}, \{3,4\}, \{1,2,3\}, \{1,2,4\},$

$\{1,3,4\}, \{2,3,4\}, \{1,2,3,4\}\}$

$B = \{\{1,2\}, \{1,3\}, \{1,4\}, \{2,3\}, \{2,4\}, \{3,4\}\}$

$\boxed{C = \{\phi, \{1\}, \{2\}, \{3\}, \{4\}\}}$

Q16) $B = \{(a, S) \in A \times P(A) : a \in S\}$

a) $A = \{1\}$; $P(A) = \{\phi, \{1\}\}$; $\boxed{B = \{(1, \{1\})\}}$

b) $A = \{1,2\}$; $P(A) = \{\phi, \{1\}, \{2\}, \{1,2\}\}$; $\boxed{B = \{(1, \{1\}), (2, \{2\}), (1, \{1,2\}), (2, \{1,2\})\}}$

c) $A = \{1,2,3\}$; $\boxed{B = \{(1, \{1\}), (1, \{1,2\}), (1, \{1,3\}, (1, \{1,2,3\}), (2, \{2\}), (2, \{1,2\}), (2, \{2,3\}), (2, \{1,2,3\}), (3, \{3\}), (3, \{1,3\}), (3, \{2,3\}), (3, \{1,2,3\})\}}$

Q18) Let U be the set containing integers from 1 to 1000

Let A, B, C & D be the sets of multiples of 2, 3, 5 & 7 in U respectively.

$|A| = 500$; $|B| = 333$; $|C| = 200$; $|D| = 142$

$|A \cap B| = 166$; $|A \cap C| = 100$; $|A \cap D| = 71$
$|B \cap C| = 66$; $|B \cap D| = 47$; $|C \cap D| = 28$
$|A \cap B \cap C| = 33$; $|A \cap B \cap D| = 23$; $|A \cap C \cap D| = 14$; $|B \cap C \cap D| = 9$
$|A \cap B \cap C \cap D| = 4$

$\therefore$ Using Inclusion-Exclusion Principle,

$$\boxed{\begin{aligned} |A \cup B \cup C \cup D| &= |A| + |B| + |C| + |D| - |A \cap B| - |A \cap C| - |A \cap D| \\ &\quad - |B \cap C| - |B \cap D| - |C \cap D| + |A \cap B \cap C| + |A \cap B \cap D| \\ &\quad + |A \cap C \cap D| + |B \cap C \cap D| - |A \cap B \cap C \cap D| \\ &= 772 \end{aligned}}$$

Alternatively, we can denote the sets A, B, C & D as M_1, M_2, M_3 & M_4. Therefore,

$$|M_1 \cup M_2 \cup M_3 \cup M_4| = \sum_{i=1}^{4} |M_i| - \sum_{1 \le i \le j} |M_i \cap M_j| + \sum_{1 \le i \le j < k} |M_i \cap M_j \cap M_k| - |M_1 \cap M_2 \cap M_3 \cap M_4|$$

$$= 1175 - 478 + 79 - 4$$

$$= 772$$